

The Math Behind This Library

The neural network is composed of many layers. There are n hidden layers denoted by $[l]$ and for simplicity at this stage we assume there is one output layer $[L]$. Starting from a perceptron, the i^{th} perceptron accepts a vector $a^{[l-1]}$ of size k which represents the output of the **entire** previous layer. The perceptron/neuron then holds one weight per value of $a^{[l-1]}$ and thus each perceptron holds a vector of weights $W_i^{[l]}$. Each perceptron also holds a bias that we introduce in training the perceptron which is a scalar value of b_i . The output z_i for the i^{th} perceptron is the following:

$$z_i^{[l]} = b_i + \sum_{j=1}^k W_{ij}^{[l]} \cdot a_j^{[l-1]} \quad (1)$$

To account for non-linear patterns, we pass the z_i value through an activation function $f(z_i)$ defined as:

$$a_i^{[l]} = f(z_i^{[l]}) \quad (2)$$

To calculate how much our output deviates we defined a loss function $L(a^{[l]})$. Now $a^{[l]}$ is a function of $z^{[l]}$ which is a function of the weights of that layer $W^{[l]}$. Thus Loss can be defined as follows:

$$L(a^{[l]}(z^{[l]}(W^{[l]}))) \quad (3)$$

To calculate how each layer's output $z^{[l]}$ contributes to the final loss, we need to calculate the error signal for that layer defined as $\delta^{[l]}$:

$$\delta^{[l]} = \frac{\partial L^{[l]}}{\partial z^{[l]}} = \left(\frac{\partial L^{[l]}}{\partial a^{[l]}} \right) \left(\frac{\partial a^{[l]}}{\partial z^{[l]}} \right) \quad (4)$$

Given the following equation:

$$a^{[l]} = f(z^{[l]}) \quad (5)$$

We can calculate $\frac{\partial a^{[l]}}{\partial z^{[l]}}$ to be:

$$\frac{\partial a^{[l]}}{\partial z^{[l]}} = f'(z^{[l]}) \quad (6)$$

Substituting that back:

$$\delta^{[l]} = \frac{\partial L^{[l]}}{\partial a^{[l]}} \odot f'(z^{[l]}) \quad (7)$$

Now how do we calculate $\frac{\partial L}{\partial a^{[l]}}$? We go back to original perceptron equation where we can see that $z^{[l+1]}$ is a function of $a^{[l]}$:

$$z^{[l+1]} = W^{[l+1]}a^{[l]} + b^{[l+1]} \quad (8)$$

$$\frac{\partial z^{[l+1]}}{\partial a^{[l]}} = W^{[l+1]} \quad (9)$$

Now re-defining Loss function to be a function of the next layer $z^{[l+1]}$ which is a function of the present layer's activated output $a^{[l]}$:

$$L(z^{[l+1]}(a^{[l]})) \quad (10)$$

$$\frac{\partial L}{\partial a^{[l]}} = \frac{\partial L}{\partial z^{[l+1]}} \frac{\partial z^{[l+1]}}{\partial a^{[l]}} \quad (11)$$

$$\implies \frac{\partial L}{\partial a^{[l]}} = (W^{[l+1]})^T \delta^{[l+1]} \quad (12)$$

Therefore, we come with a final expression for the error signal of a layer to be:

$$\delta^{[l]} = \frac{\partial L}{\partial z^{[l]}} = (W^{[l+1]})^T \delta^{[l+1]} \odot f'(z^{[l]}) \quad (13)$$

Moving back to the loss function, to calculate the weight gradient, we use the following chain:

$$\frac{\partial L}{\partial W^{[l]}} = \left(\frac{\partial L}{\partial a^{[l]}} \right) \left(\frac{\partial a^{[l]}}{\partial z^{[l]}} \right) \left(\frac{\partial z^{[l]}}{\partial W^{[l]}} \right) \quad (14)$$

Substituting $\delta^{[l]}$ into this, we get:

$$\frac{\partial L}{\partial W^{[l]}} = \delta^{[l]} \left(\frac{\partial z^{[l]}}{\partial W^{[l]}} \right) \quad (15)$$

Given the following:

$$z^{[l]} = W^{[l]} \cdot a^{[l-1]} + b^{[l]} \quad (16)$$

$$\frac{\partial z^{[l]}}{\partial W^{[l]}} = a^{[l-1]} \quad (17)$$

Substituting that into our gradient loss equation, we get:

$$\frac{\partial L}{\partial W^{[l]}} = \delta^{[l]} (a^{[l-1]})^T \quad (18)$$

Do not forget the bias!!! We can write the Loss Function as:

$$L(z^{[l]}(b^{[l]})) \quad (19)$$

To calculate the bias gradient, calculate the following:

$$\frac{\partial L}{\partial b^{[l]}} = \left(\frac{\partial L}{\partial z^{[l]}} \right) \left(\frac{\partial z^{[l]}}{\partial b^{[l]}} \right) \quad (20)$$

$$\frac{\partial L}{\partial b^{[l]}} = \delta^{[l]} \left(\frac{\partial z^{[l]}}{\partial b^{[l]}} \right) \quad (21)$$

$$z^{[l]} = W^{[l]} \cdot a^{[l-1]} + b^{[l]}, \quad \frac{\partial z^{[l]}}{\partial b^{[l]}} = 1 \quad (22)$$

Therefore,

$$\frac{\partial L}{\partial b^{[l]}} = \delta^{[l]} \quad (23)$$